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Multipliers of Fourier transforms and series on L^1

By

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1. Introduction. Let $\mathscr{L} := L^1(-\infty, \infty)$ be the space of all complex-valued functions f that are integrable in Lebesgue's sense over the real line and endowed with the norm

$$\|f\|_{\mathscr{L}} := \int_{-\infty}^{\infty} |f(x)|\,dx.$$

Denote by $\hat{\mathscr{L}}$ the space of the Fourier transforms $\hat{f}$ of the functions $f \in \mathscr{L}$ defined by

$$\hat{f}(t) := \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-itx}\,dx, \quad -\infty < t < \infty. \tag{1.1}$$

We say that a function λ measurable and bounded on $(-\infty, \infty)$ is an $(\mathscr{L} \to \mathscr{L})$ multiplier, or briefly a multiplier on $\mathscr{L}$, if the product $\lambda \hat{f}$ belongs to $\hat{\mathscr{L}}$ for every $f \in \mathscr{L}$. In other words, λ is a multiplier if for every $f \in \mathscr{L}$ there exists a function $g \in \mathscr{L}$ such that

$$\lambda(t)\hat{f}(t) = \hat{g}(t), \quad -\infty < t < \infty. \tag{1.2}$$

Let A be defined as the operator of $\mathscr{L}$ into $\mathscr{L}$ which assigns to each $f \in \mathscr{L}$ the function $A(f) := g$, where $g \in \mathscr{L}$ is given by (1.2). It is plain that $A(f)$ is linear and closed. Thus, by the closed graph theorem, $A(f)$ is a bounded operator in any case.

Denote by $\mathscr{B}$ the space of the functions λ that are absolutely continuous and bounded over the real line, and endowed with the norm

$$\|\lambda\|_{\mathscr{B}} := \sup|\lambda(t)| + \int_{-\infty}^{\infty} |\lambda'(t)|\,dt,$$

where the supremum is extended over $t \in (-\infty, \infty)$. As is well-known, there exists a function λ in $\mathscr{B}$ which is not a multiplier (see, e.g. [8, pp. 170–172]).

2. Multipliers of Fourier transforms. We introduce a family $\{\mathscr{B}_p : 1 \leqq p < \infty\}$ of subspaces of $\mathscr{B}$, whose elements are multipliers of Fourier transforms on $\mathscr{L}$. We define $\mathscr{B}_p$

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for $1 < p < \infty$ to be

$$\mathscr{B}_p := \left\{\lambda: \|\lambda\|_{\mathscr{B}_p} := \sup|\lambda(t)| + \int_0^\infty \left(\frac{1}{u}\int_{u\leqq|t|\leqq 2u} |\lambda'(t)|^q\,dt\right)^{1/q} du + \int_0^\infty \frac{|\lambda(t)-\lambda(-t)|}{t}\,dt < \infty\right\},$$

where q is the conjugate exponent to p, i.e. $1/p + 1/q = 1$; while for $p = 1$ (this time $q := \infty$) to be

$$\mathscr{B}_1 := \left\{\lambda: \|\lambda\|_{\mathscr{B}_1} := \sup|\lambda(t)| + \int_0^\infty (\operatorname{ess\,sup}_{u\leqq|t|\leqq 2u} |\lambda'(t)|)\,du + \int_0^\infty \frac{|\lambda(t)-\lambda(-t)|}{t}\,dt < \infty\right\}.$$

Now note that if the term $\sup|\lambda(t)|$ is replaced by $|\lambda(0)|$, then we get another norm equivalent to the one defined above.

By Fubini's theorem,

$$\int_0^\infty \frac{1}{u}\int_{u\leqq|t|\leqq 2u} |\lambda'(t)|\,dt\,du = (\ln 2)\int_{-\infty}^\infty |\lambda'(t)|\,dt$$

and by Hölder's inequality,

$$\int_0^\infty \left(\frac{1}{2u}\int_{u\leqq|t|\leqq 2u} |\lambda'(t)|^{q_1}\,dt\right)^{1/q_1} du \leqq \int_0^\infty \left(\frac{1}{2u}\int_{u\leqq|t|\leqq 2u} |\lambda'(t)|^{q_2}\,dt\right)^{1/q_2} du, \quad 0 < q_1 < q_2 < \infty.$$

Hence it follows that

$$\mathscr{B}_1 \subseteqq \mathscr{B}_{p_1} \subseteqq \mathscr{B}_{p_2} \subseteqq \mathscr{B}, \quad 1 < p_1 < p_2 < \infty.$$

Our first main result is the following

Theorem 1. *If a function λ belongs to $\mathscr{B}_p$ for some $1 \leqq p < \infty$, then λ is a multiplier of Fourier transform on $\mathscr{L}$ and for every $f \in \mathscr{L}$ we have*

$$\|\Lambda(f)\|_{\mathscr{L}} \leqq C_p \|\lambda\|_{\mathscr{B}_p} \|f\|_{\mathscr{L}}. \tag{2.1}$$

Here and in the sequel, by C_p we denote a constant depending only on p, which is not necessarily the same at different occurrencies.

As is known (see, e.g. [3, p. 269]), a necessary and sufficient condition for a function λ to be a multiplier on $\mathscr{L}$ is that λ is the Fourier-Stieltjes transform of a function μ of bounded variation. We remind the reader that the Fourier-Stieltjes transform $\hat{\mu}$ of such μ is defined by

$$\hat{\mu}(t) := \frac{1}{2\pi}\int_{-\infty}^\infty e^{-itx}\,d\mu(x), \quad -\infty < t < \infty.$$

In particular, Bochner [1] (see also [3, p. 241]) proved that a necessary and sufficient condition for representation of λ in terms of a Fourier-Stieltjes transform of a bounded, nondecreasing function is that λ be positive semidefinite. However, it is difficult to verify the fulfilment of this property in general.

Butzer and Nessel [3, p. 248] gave a sufficient condition, which is easier to check in practice. They call a function λ, defined on $[0, \infty]$, quasi-convex if it is locally absolutely continuous, its derivative λ' is of bounded variation, and such that

$$\int_0^\infty t\,|d\lambda'(t)| < \infty\,.$$

Now, the following theorem is proved in [3, pp. 251–252]: If a function λ is even, quasi-convex, and such that

$$\lim_{|t| \to \infty} \lambda(t) = 0, \tag{2.2}$$

then λ is representable as the Fourier transform of an even function in $\mathscr{L}$. In particular, $\lambda \in \mathscr{L}$.

Our Theorem 1 also implies this result. Indeed, we may write

$$\lambda'(t) = h_1(t) - h_2(t),$$

where h_1 and h_2 are nonnegative and nonincreasing functions on $[0, \infty)$ and such that

$$\int_0^\infty t\,dh_j(t) > -\infty \quad (j = 1, 2).$$

By integrating by parts, hence it follows that

$$\begin{aligned} \mathscr{A}_\infty &:= \int_0^\infty (\operatorname{ess\,sup}_{u \leqq |t| \leqq 2u} |\lambda'(t)|)\,du \\ &\leqq 2 \int_0^\infty (h_1(t) + h_2(t))\,dt \\ &= \sum_{j=1}^{2} \int_0^\infty t\,d(-h_j(t)) < \infty\,. \end{aligned} \tag{2.3}$$

This means that $\lambda \in \mathscr{B}_1$ and our Theorem 1 applies.

Finally, we present an even function $\lambda \in \mathscr{B}_2$, which is not quasi-convex.

Example. Let

$$\lambda'(t) := \begin{cases} [2^{m/2}\, m \ln^2 (m+1)]^{-1} & \text{if} \quad t = 2^m, m = 1, 2, \ldots, \\ \text{linear on the intervals } [2^m - 1, 2^m] & \text{and} \quad [2^m, 2^m + 1], \\ 0 & \text{elsewhere on} \quad [0, \infty), \\ -\lambda'(-t) & \text{on} \quad (-\infty, 0], \end{cases}$$

and define λ to satisfy condition (2.2). Then λ is even and

$$\mathscr{A}_\infty = 2\sum_{m=1}^{\infty}\int_{2^{m-1}}^{2^m}\Big(\max_{u\leqq|t|\leqq 2u}|\lambda'(t)|\Big)\,du$$
$$\geqq \sum_{m=1}^{\infty}\frac{2^{m/2}}{m\ln^2(m+1)} = \infty .$$

This shows that λ cannot be quasi-convex (cf. (2.3)).

On the other hand,

$$\sum_{m=1}^{\infty}\int_{2^{m-1}}^{2^m}\left(\frac{1}{u}\int_{u}^{2u}|\lambda'(t)|^2\,dt\right)^{1/2}du \leqq \sum_{m=1}^{\infty}\int_{2^{m-1}}^{2^m}2^{-(m-1)/2}\left(\int_{2^{m-1}}^{2^{m+1}}|\lambda'(t)|^2\,dt\right)^{1/2}du$$
$$= \sum_{m=1}^{\infty}2^{(m-1)/2}\left(\int_{2^{m-1}}^{2^{m+1}}|\lambda'(t)|^2\,dt\right)^{1/2}du$$
$$\leqq \sqrt{3}\sum_{m=1}^{\infty}[m\ln^2(m+1)]^{-1} < \infty .$$

This means that $\lambda\in\mathscr{B}_2$ and our Theorem 1 applies.

3. Proof of Theorem 1. It hinges on the following two results.

Lemma 1 (see [4]). *If a function λ is locally absolutely continuous, satisfies condition* (2.2), *and for some $1 < q \leqq \infty$ we have*

$$\mathscr{A}_q := \int_0^{\infty}\left(\frac{1}{u}\int_{u\leqq|x|\leqq 2u}|\lambda'(t)|^q\,dt\right)^{1/q}du < \infty ,$$

then the function $\hat{\lambda}$ defined by

$$\hat{\lambda}(x) := \lim_{T\to\infty}\frac{1}{2\pi}\int_{-T}^{T}\lambda(t)e^{-itx}\,dt, \quad -\infty < x < \infty , \tag{3.1}$$

belongs to $\mathscr{L}$ if and only if

$$\mathscr{A} := \int_0^{\infty}\frac{|\lambda(t)-\lambda(-t)|}{t}\,dt < \infty ,$$

in which case the inversion formula

$$\lambda(t) = \int_{-\infty}^{\infty}\hat{\lambda}(x)e^{ixt}\,dx \tag{3.2}$$

holds. Furthermore,

$$\|\hat{\lambda}\|_{\mathscr{L}} \leqq \mathscr{A} + C_q\mathscr{A}_q . \tag{3.3}$$

We remind the reader that the convolution of two functions f and g is defined by

$$(f*g)(x) := \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x-t)\,g(t)\,dt, \quad -\infty < x < \infty.$$

Lemma 2 (see, e.g. [3, pp. 5 and 189]). *If $f \in \mathscr{L}$ and $g \in \mathscr{L}$, then*

$$\|f*g\|_{\mathscr{L}} \leqq \|f\|_{\mathscr{L}}\,\|g\|_{\mathscr{L}} \tag{3.4}$$

and

$$(f*g)^{\wedge}(t) = \hat{f}(t)\,\hat{g}(t), \quad -\infty < t < \infty. \tag{3.5}$$

After these preliminaries, we turn to the proof of Theorem 1. To this effect, let $\lambda \in \mathscr{B}_p$ for some $1 \leqq p < \infty$. From the fact that $\lambda' \in \mathscr{L}$ it follows easily that the finite limits

$$\lim_{t\to\infty} \lambda(t) = l_1 \quad \text{and} \quad \lim_{t\to-\infty} \lambda(t) = l_2$$

exist. By virtue of the fact that

$$\int_0^{\infty} \frac{|\lambda(t)-\lambda(-t)|}{t}\,dt < \infty, \tag{3.6}$$

we have necessarily that $l_1 = l_2 =: l$. In fact, in case $l_1 \neq l_2$ we may conclude that

$$\begin{aligned} |\lambda(t)-\lambda(-t)| &\geqq |l_1-l_2| - |\lambda(t)-l_1| - |\lambda(-t)-l_2| \\ &\geqq \tfrac{1}{2}|l_1-l_2| \quad \text{for all large enough } t, \end{aligned}$$

which contradicts (3.6). Thus, the finite limit

$$\lim_{|t|\to\infty} \lambda(t) = l$$

exists. Define new functions λ_0 and λ_1 as follows

$$\lambda_0(t) := \lambda(t) - l \quad \text{and} \quad \lambda_1(t) := 2\pi\hat{\lambda}_0(-t), \quad -\infty < t < \infty, \tag{3.7}$$

where $\hat{\lambda}_0$ is defined accordingly to (3.1). By Lemma 1, $\lambda_1 \in \mathscr{L}$. Thus, by (1.1) and (3.2) we have

$$\hat{\lambda}_1(t) = \lambda_0(t). \tag{3.8}$$

Now, we claim that

$$\Lambda(f) = lf + \lambda_1 * f, \quad f \in \mathscr{L}.$$

Indeed, by (3.5), (3.7), and (3.8),

$$\begin{aligned} [\Lambda(f)]^{\wedge}(t) &= l\hat{f}(t) + \hat{\lambda}_1(t)\,\hat{f}(t) \\ &= l\hat{f}(t) + \lambda_0(t)\,\hat{f}(t) \\ &= \lambda(t)\,\hat{f}(t), \quad -\infty < t < \infty. \end{aligned}$$

This proves that λ is a multiplier on $\mathscr{L}$.

By the definition of the norm $\|\cdot\|_{\mathscr{B}_p}$, (3.3), and (3.4),

$$\|\Lambda(f)\|_{\mathscr{L}} \leqq |l|\,\|f\|_{\mathscr{L}} + \|\lambda_1\|_{\mathscr{L}}\,\|f\|_{\mathscr{L}} \leqq C_q\,\|\lambda\|_{\mathscr{B}_p}\,\|f\|_{\mathscr{L}}, \quad 1/p + 1/q = 1.$$

This proves (2.1).

4. Multipliers of Fourier series. We observe that such multipliers can be obtained in the same manner as in the case of Fourier transforms. We briefly summarize the relevant notions, results, and reasoning.

Let $L = L^1(-\pi, \pi)$ be the space of all complex-valued, 2π-periodic functions f that are integrable in Lebesgue's sense over the interval $(-\pi, \pi)$ and endowed with the norm

$$\|f\|_L := \int_{-\pi}^{\pi} |f(x)|\,dx.$$

Denote by $\hat{L}$ the space of the sequences of Fourier coefficients

$$\hat{f}(k) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,e^{-ikx}\,dx, \quad k = 0, \pm 1, \pm 2, \ldots,$$

of the functions $f \in L$.

We say that a bounded sequence $\{\lambda = \lambda(k)\}$ is an $(L \to L)$ multiplier, or briefly a multiplier on L, if the product sequence $\{\lambda(k)\,\hat{f}(k)\}$ belongs to $\hat{L}$ for every $f \in L$. In other words, λ is a multiplier if for every $f \in L$ there exists a function $g \in L$ such that

$$\lambda(k)\,\hat{f}(k) = \hat{g}(k), \quad k = 0, \pm 1, \pm 2, \ldots. \tag{4.1}$$

Let Λ be defined as the operator of L into L which assigns to each $f \in L$ the function $\Lambda(f) := g$, where $g \in L$ is given by (4.1). Clearly, $\Lambda(f)$ is a bounded linear operator.

We remind the reader that the space of the sequences of bounded variation is defined by

$$bv := \left\{\lambda: \|\lambda\|_{bv} := \sup|\lambda(k)| + \sum_{k=-\infty}^{\infty} |\Delta\lambda(k)| < \infty\right\},$$

where the supremum is extended over $k = 0, \pm 1, \pm 2, \ldots$ and the difference operator Δ is defined by

$$\Delta\lambda(k) := \begin{cases} \lambda(k) - \lambda(k+1) & \text{if} \quad k = 0, 1, 2, \ldots, \\ \lambda(k) - \lambda(k-1) & \text{if} \quad k = -1, -2, \ldots. \end{cases}$$

As is well-known, there exists a sequence λ in bv which is not a multiplier (see, e.g. [9, Vol. 1, p. 184]).

We introduce a family $\{bv_p: 1 \leqq p < \infty\}$ of subspaces of bv whose elements are multipliers of Fourier series on L. We define bv_p for $1 < p < \infty$ to be

$$bv_p := \left\{\lambda: \|\lambda\|_{bv_p} := \sup|\lambda(k)| + \sum_{n=0}^{\infty} 2^n\left(2^{-n}\sum_{|k|\in I_n} |\Delta\lambda(k)|^q\right)^{1/q} + \sum_{k=1}^{\infty}\frac{|\lambda(k) - \lambda(-k)|}{k} < \infty\right\},$$

where $I_n := \{2^n, 2^n + 1, \dots, 2^{n+1} - 1\}$ for $n = 0, 1, \dots$ and $1/p + 1/q = 1$; while for $p = 1$ (this time $q := \infty$) to be

$$b v_1 := \left\{ \lambda : \| \lambda \|_{b v_1} := \sup |\lambda(k)| + \sum_{n=0}^{\infty} 2^n \max_{|k| \in I_n} |\Delta \lambda(k)| + \sum_{k=1}^{\infty} \frac{|\lambda(k) - \lambda(-k)|}{k} < \infty \right\}.$$

Again, if the term $|\lambda(0)|$ is substituted for $\sup |\lambda(k)|$, then we get another norm equivalent to the one defined above.

By Hölder's inequality, it is not difficult to check that

$$b v_1 \subseteq b v_{p_1} \subseteq b v_{p_2} \subseteq b v, \quad 1 < p_1 < p_2 < \infty.$$

Our second main result is the following.

Theorem 2. *If a sequence λ belongs to $b v_p$ for some $1 \leqq p < \infty$, then λ is a multiplier of Fourier series on L and for every $f \in L$ we have*

$$\| A(f) \|_L \leqq C_p \| \lambda \|_{b v_p} \| f \|_L. \tag{4.2}$$

We note that Theorem 2 in the particular case of even sequences λ, that is when

$$\lambda(-k) = \lambda(k), \quad k = 1, 2, \dots,$$

was proved by Buntinas [2], while giving a direct proof through the strong summability of Fourier series. He also points out that this particular case can be derived from the integrability results of Fomin [5].

As is known (see, e.g. [3, p. 268]), a necessary and sufficient condition for a sequence λ to be a multiplier on L is that λ can be represented as the sequence of the Fourier-Stieltjes coefficients of a 2π-periodic function μ of bounded variation over $[-\pi, \pi]$:

$$\lambda(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikx} d\mu(x), \quad k = 0, \pm 1, \pm 2, \dots.$$

In particular, our Theorem 2 also gives an applicable criterion to the trigonometric moment problem.

The proof of Theorem 2 relies on the following two results.

Lemma 3 (see [6, 7]). *If a sequence $\{c_k : k = 0, \pm 1, \pm 2, \dots\}$ is such that*

$$\lim_{|k| \to \infty} c_k = 0$$

and for some $1 < q \leqq \infty$ we have

$$\mathscr{C}_q := \sum_{n=0}^{\infty} 2^n \left(2^{-n} \sum_{|k| \in I_n} |\Delta c_k|^q \right)^{1/q} < \infty,$$

then the sum $f=f(x)$ of the series

$$\sum_{k=-\infty}^{\infty} c_k e^{ikx} \tag{4.3}$$

belongs to L if and only if

$$\mathscr{C} := \sum_{k=1}^{\infty} \frac{|c_k - c_{-k}|}{k} < \infty,$$

in which case (4.3) *is the Fourier series of its sum f. Furthermore,*

$$\| f \|_L \leqq \mathscr{C} + C_q \mathscr{C}_q.$$

The necessary and sufficient part of Lemma 3 is due to Fournier and Self [6], while the last two statements were proved by the second named author in [7], where a different proof of the whole lemma is presented. We note that Lemma 3 can also be deduced from Lemma 1.

Finally, we need the 2π-periodic version of Lemma 2, when the convolution of the 2π-periodic functions f and g is defined by

$$(f*g)(x) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) g(t)\, dt, \quad -\pi \leqq x < \pi.$$

Lemma 4 (see, e.g. [9, Vol. 1, pp. 36–37]). *If $f \in L$ and $g \in L$, then*

$$\| f*g \|_L \leqq \| f \|_L \| g \|_L$$

and

$$(f*g)^{\wedge}(k) = \hat{f}(k)\hat{g}(k), \quad k = 0, \pm 1, \ldots.$$

The proof of Theorem 2 runs along the same lines as that of Theorem 1. Therefore, we only sketch it.

From the fact that $\lambda \in bv_p$ for some $1 \leqq p < \infty$ it follows that the finite limit

$$\lim_{|k| \to \infty} \lambda(k) = l$$

exists. We define a new sequence λ_0 by

$$\lambda_0(k) := \lambda(k) - l, \quad k = 0, \pm 1, \ldots.$$

Then $\lambda_0 \in \hat{L}$ and the function λ_1 defined by

$$\lambda_1(x) := \sum_{k=-\infty}^{\infty} \lambda_0(k) e^{ikx}, \quad -\pi \leqq x < \pi,$$

belongs to L and $\hat{\lambda}_1 = \lambda_0$. We set

$$\Lambda(f) := lf + \lambda_1 * f, \quad f \in L,$$

and proceed exactly in the same way as in the proof of Theorem 1. As a result, we find that λ is a multiplier on L and (4.2) is also satisfied. This completes the proof of Theorem 2.

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